

Static Website Deployment to S3 using GitHub Actions

Introduction

This project demonstrates how to deploy a static website to an AWS S3 bucket using GitHub Actions. The website files are stored in a directory and automatically pushed to S3 on each GitHub commit.

Abstract

The goal is to automate the deployment process of a static website using GitHub Actions. This removes the need for manual uploads to S3 and ensures the latest code is always live.

Tools Used

- GitHub
- AWS S3
- GitHub Actions
- Static Website Files (HTML/CSS/JS)

Steps Involved in Building the Project

1. Create an S3 bucket with static website hosting enabled.
2. Make the bucket public by updating the bucket policy.
3. Prepare your static website files (HTML, CSS, JS).
4. Organize your files in a directory (e.g., 'Project-1').
5. Push the project to a GitHub repository.
6. Create a GitHub Actions workflow YAML file in ``.github/workflows/deploy.yml``.
7. In the workflow file, configure AWS credentials using secrets and sync files to S3 using ``aws s3 sync ./Project-1 s3://your-bucket-name``.
8. Commit and push the workflow file to trigger deployment.

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9. Access the website using the S3 public URL:<https://s3-with-github-action.s3.amazonaws.com/index.html>

Conclusion

Using GitHub Actions and AWS S3 together creates a powerful and efficient deployment pipeline. It ensures that updates to your static website are automatically reflected live, making development and maintenance smoother.